Machine Learning for Health Monitoring Using Wearable Devices:

Applications, Challenges, and Future Directions

Mohammed Adnan | Student Number: 20806056 | MSc Computer Science (AI) 2-Year Enhanced Programme | University of Nottingham

COMP4037 Research Methods | Coursework 3

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1. Introduction and Motivation

Wearable devices like smartwatches, Electrocardiogram patches and biosensor wristbands have emerged as an important element in modern healthcare. With the use of machine learning and deep learning models, these devices can diagnose diseases early, monitor chronic patients remotely, predict falls and assess mental states like stress. This area gained tremendous interest following the COVID-19 pandemic which highlighted the importance of remote health monitoring that does not depend on hospital visits (Sabry et al., 2022).

This review examines 10 publications covering cardiac monitoring, mental health, sleep staging, activity tracking, fall detection and remote patient monitoring relevant to my planned MSc dissertation at University of Nottingham.

Table 1: Metadata Summary of the 10 Reviewed Papers

No	First Author	Year	Category	Evaluation
1	Gedam & Paul	2021	Mental Health & Sleep	Lit. review
2	Huda et al.	2020	Cardiac Monitoring	Quantitative / F1
3	Islam et al.	2023	Remote Monitoring	Quantitative
4	Li et al.	2020	Activity & Fall	Quantitative
5	Mohammad et al.	2023	Activity & Fall	Quantitative
6	Prieto-Avalos et al.	2022	Cardiac Monitoring	Comparative review
7	Radha et al.	2021	Mental Health & Sleep	Quantitative / kappa
8	Sabry et al.	2022	Survey & Overview	Lit. review
9	Shaik et al.	2023	Remote Monitoring	Lit. review
10	Shajari et al.	2023	Survey & Overview	Lit. review

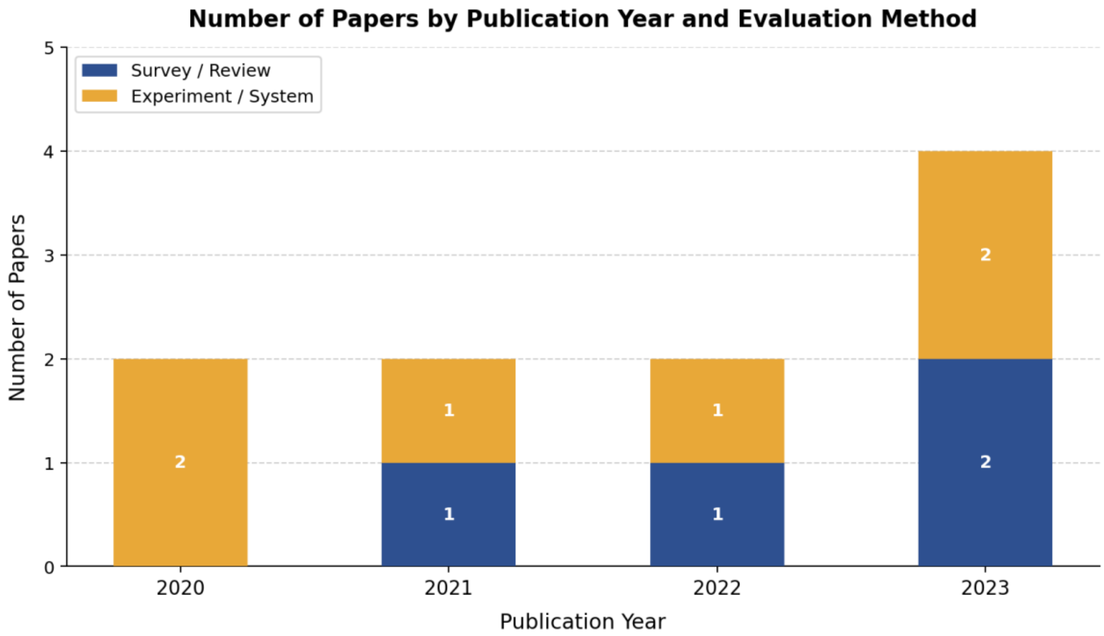


Figure 1: Number of Papers by Publication Year and Evaluation Method Available at: [https://colab.research.google.com/drive/17UL7DlvtqsFNDN5Z_ppl46GtI1H1rlZ6?usp=sharing]

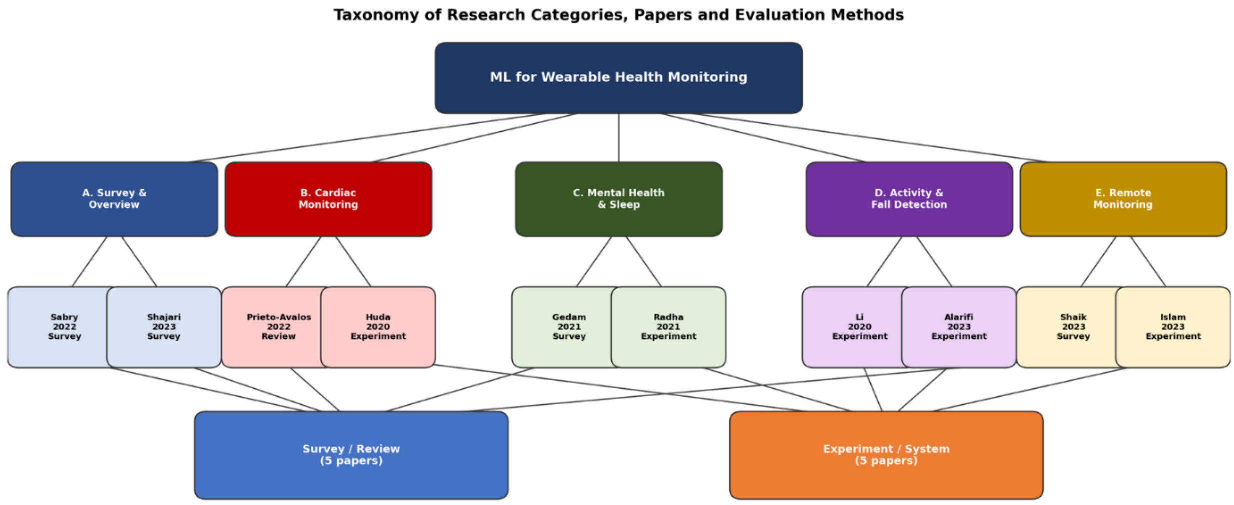


Figure 2: Taxonomy of Research Categories, Papers and Evaluation Methods. Available at: [https://colab.research.google.com/drive/17UL7DlvtqsFNDN5Z_ppl46GtI1H1rlZ6?usp=sharing]

1.1 Research Field Challenges

The researchers involved in machine learning for wearable health monitoring encounter a number of important obstacles which makes this a particularly challenging area of investigation. The four main challenges discussed in the literature review include:

- **Data Quality and Scarcity:** Clinical wearable data tends to be small scale, unbalanced and acquired under artificial laboratory conditions which affect its capacity to generalize to unseen patients (Sabry et al., 2022).

- **Cross-User Generalization:** Human bodies differ due to variations in age, weight, and fitness. This difference makes it difficult for a single model to learn across users (Gedam & Paul, 2021).
- **Power Consumption and On-Device Deployment:** Wearables have limited battery capacity and compute power, and this makes it difficult to implement deep learning models locally. This requires the use of cloud servers, but there are latency issues with this method (Sabry et al., 2022).
- **Privacy and Security:** Real time physiological monitoring brings up privacy concerns. In addition, health data from wearables falls under GDPR jurisdiction, making privacy preserving machine learning more important (Shajari et al., 2023).

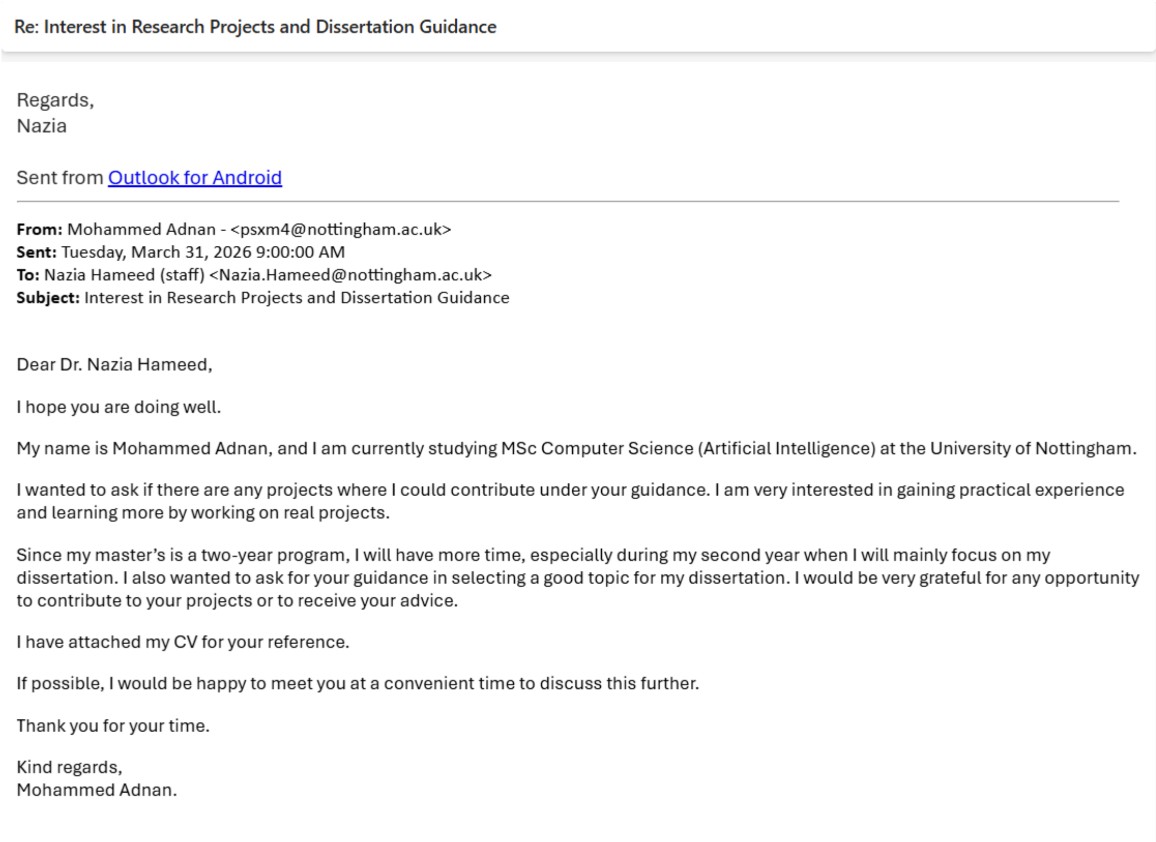
1.2 Survey Scope

In this review, we examine machine learning and deep learning techniques using wearables' data for health applications, such as cardiac monitoring, mental health monitoring, sleep analysis, activity recognition, fall detection, and telemedicine via the Internet of Things. This review includes both survey articles and experiments. Research related to implantable devices, fixed medical devices in hospitals, genomic monitoring, hardware only sensing, and smartphone only applications is not considered.

1.3 Search Methodology and Contact with Supervisor

Papers were found through Google Scholar, IEEE Xplore, PubMed and Multidisciplinary Digital Publishing Institute using keywords: 'machine learning wearable health monitoring', 'deep learning Electrocardiogram arrhythmia', 'fall detection accelerometer deep learning', 'stress detection wearable machine learning', 'sleep stage Photoplethysmography classification' and 'remote patient monitoring Internet of Things machine learning'. Papers were selected based on topic relevance, open access availability, Digital Object Identifier and venue quality. The survey by Sabry et al. (2022) was the starting point and ConnectedPapers.com was used to explore related work in the same research cluster.

As a 2-year enhanced Master of Science student i contacted Dr. Nazia Hameed, Assistant Professor in Computer Science at University of Nottingham, by email on 31 March 2026. She conducts research in mobile-enabled melanoma skin cancer detection using Support Vector Machine which sparked my interest in machine learning for medical applications. Dr. Hameed arranged an online Microsoft Teams meeting on 1 April 2026 where we discussed the topic and she suggested reading a survey paper to better understand the field.



Re: Interest in Research Projects and Dissertation Guidance

Regards,
Nazia

Sent from [Outlook for Android]

From: Mohammed Adnan - <psxm4@nottingham.ac.uk>
Sent: Tuesday, March 31, 2026 9:00:00 AM
To: Nazia Hameed (staff) <Nazia.Hameed@nottingham.ac.uk>
Subject: Interest in Research Projects and Dissertation Guidance

Dear Dr. Nazia Hameed,

I hope you are doing well.

My name is Mohammed Adnan, and I am currently studying MSc Computer Science (Artificial Intelligence) at the University of Nottingham.

I wanted to ask if there are any projects where I could contribute under your guidance. I am very interested in gaining practical experience and learning more by working on real projects.

Since my master's is a two-year program, I will have more time, especially during my second year when I will mainly focus on my dissertation. I also wanted to ask for your guidance in selecting a good topic for my dissertation. I would be very grateful for any opportunity to contribute to your projects or to receive your advice.

I have attached my CV for your reference.

If possible, I would be happy to meet you at a convenient time to discuss this further.

Thank you for your time.

Kind regards,
Mohammed Adnan.

Figure 3: Email sent to Dr. Nazia Hameed.

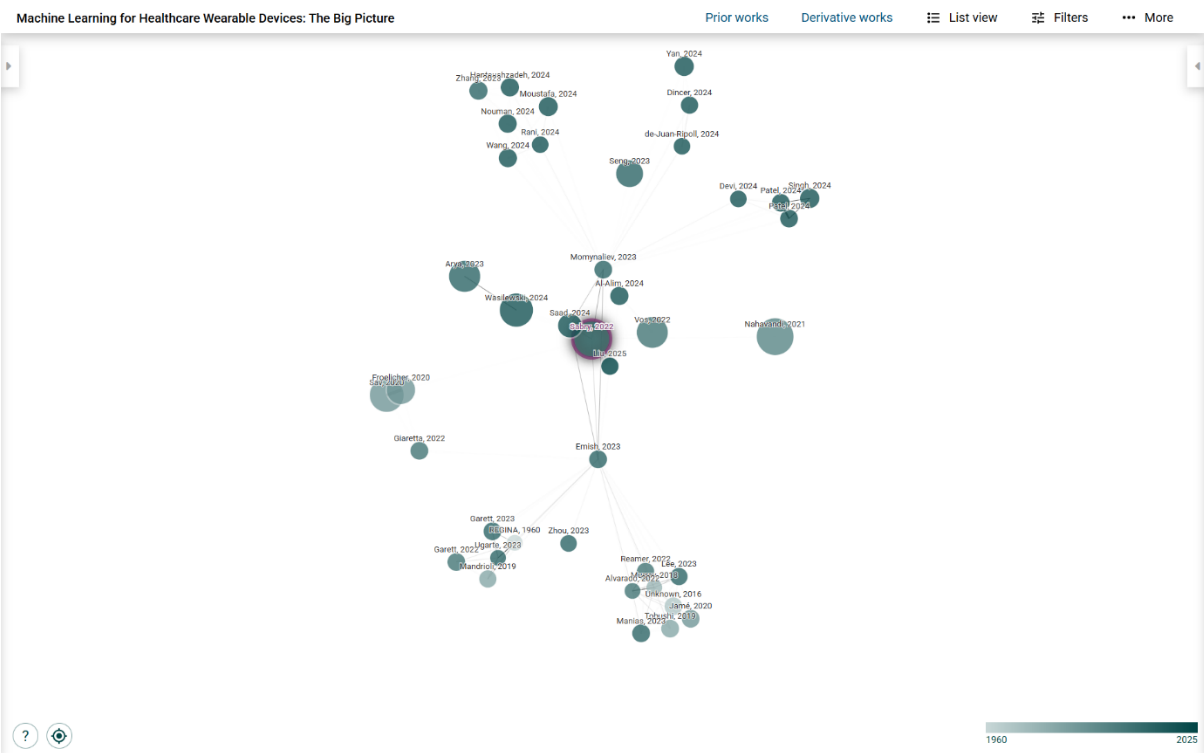


*Figure 4: [ConnectedPapers.com] graph centred on [Sabry et al. (2022),]
Available at:
[https://www.connectedpapers.com/main/b97bf4220985e0a2e95f8fe01be5f11ea7a38166/Machine-Learning-for-Healthcare-Wearable-Devices%3A-The-Big-Picture/graph]*

1.4 Classification of Literature and Organisation

The 10 articles have been classified into 5 groups with 2 articles in each group. Group A consists of two general review articles. Group B includes two articles on cardiac monitoring. Group C includes two articles related to mental health and sleep disorders. Group D includes two articles about activity recognition and fall detection. Lastly, Group E includes two articles on remote patient monitoring.

The SurVis interactive bibliography is available at:
https://mohammedadnan04.github.io/survis/src/index.html

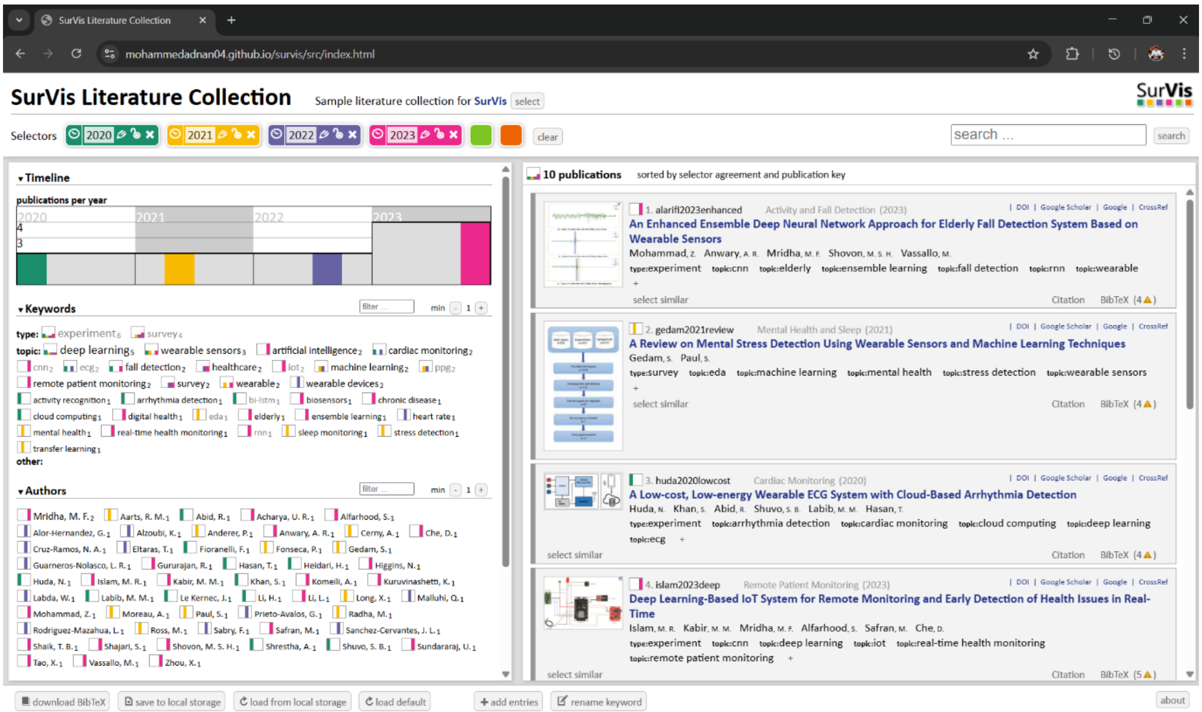


Figure 5: SurVis screenshot showing all 10 papers with required filters. Click on the items to set filter [https://mohammedadnan04.github.io/survis/src/index.html]

2. Paper Summaries

This section contains brief descriptions of each of the 10 chosen papers that fall under the 5 classifications set out in Section 1.4. The sections start from a general overview of the literature, move on to the topic of heart health, mental well being and sleep, physical activity and fall detection, and end with remote monitoring systems. Every paper features an illustrative diagram.

2.1 Category A: Survey and Overview Papers

(Paper 1) [Sabry et al. (2022)]() presents a broad survey of machine learning and deep learning applied to wearable healthcare devices, covering fall detection, activity recognition, stress, arrhythmia and sleep monitoring. We observe in Figure 6 a useful taxonomy that maps sensor signals Electrocardiogram, Photoplethysmography, Electroencephalogram, accelerometer and Electrodermal Activity to specific machine learning tasks. A structured literature review across 2017 to 2021 publications finds that deep learning outperforms classical methods like Support Vector Machine in median accuracy, which signifies how much the field has progressed in recent years. Key challenges identified include data scarcity, power consumption and privacy which are still major unsolved problems.

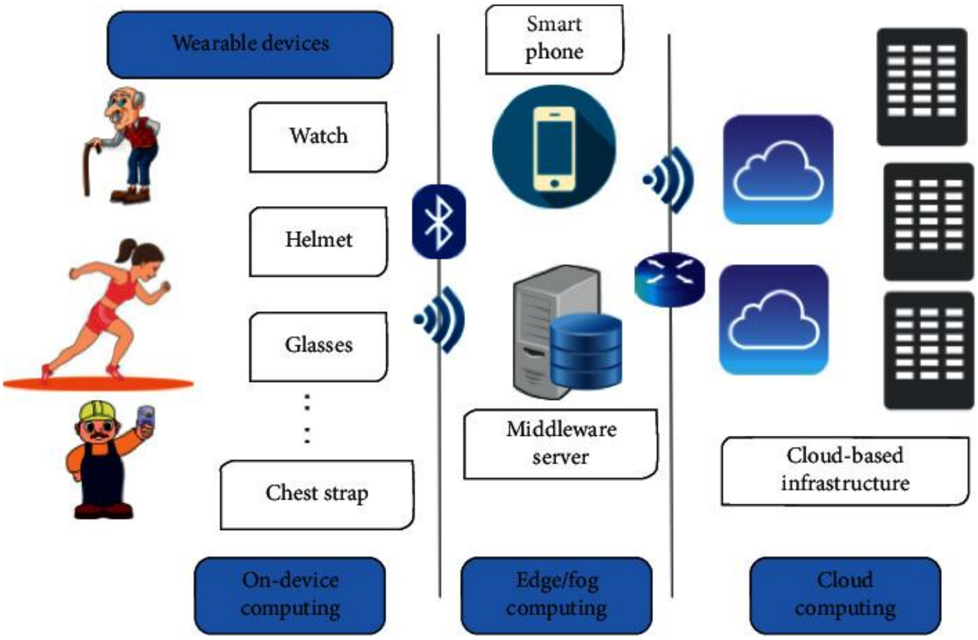


Figure 6: Wearable device application model ([Sabry et al., 2022](#)).

(Paper 2) [Shajari et al. (2023)](#) takes a broader view by examining Artificial Intelligence based wearable sensors alongside their algorithms, covering physical, chemical and biosensors, and argues that both hardware and algorithms must advance together for this field to progress. Devices are categorised into three groups and Artificial Intelligence applications are reviewed for each as we come across in Figure 7. Based on a narrative review of over 200 works, a major finding is that very few Artificial Intelligence powered wearable systems have reached commercial deployment, which signifies the gap that araises between research performance and real world reliability.

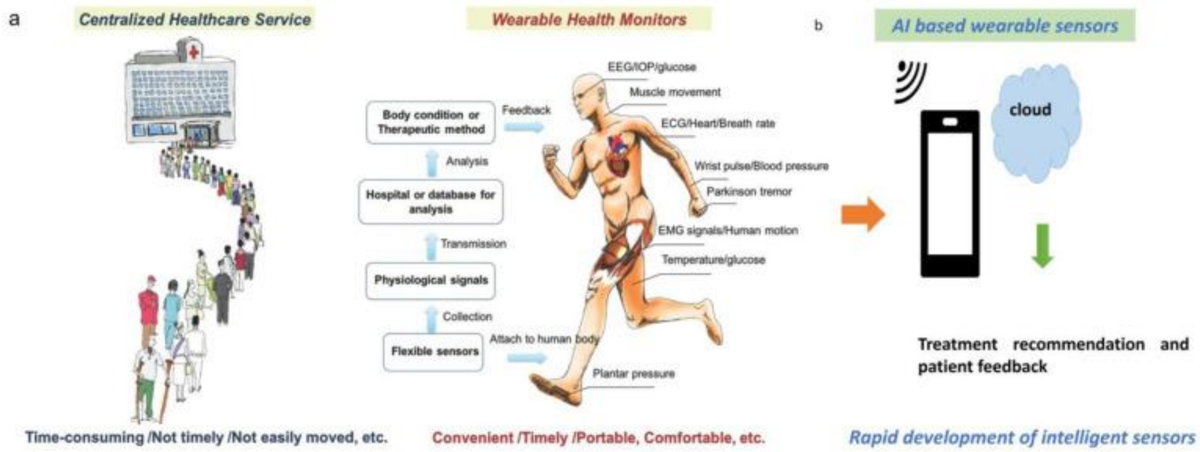


Figure 7: AI-Enabled Personalised Healthcare and Monitoring diagram ([Shajari et al., 2023](#)).

2.2 Category B: Cardiac Monitoring

(Paper 3) [Prieto-Avalos et al. (2022)](#) provides a systematic review of commercial and non-commercial wearable devices for cardiovascular monitoring, comparing Photoplethysmography based consumer smartwatches against Electrocardiogram based research prototypes as shown in Figure 8. The comparative literature review analysing accuracy, energy consumption and cost finds that commercial wearables are suitable for general wellness monitoring but insufficient for serious cardiac diagnosis, which is a significant limitation we come across in this area.

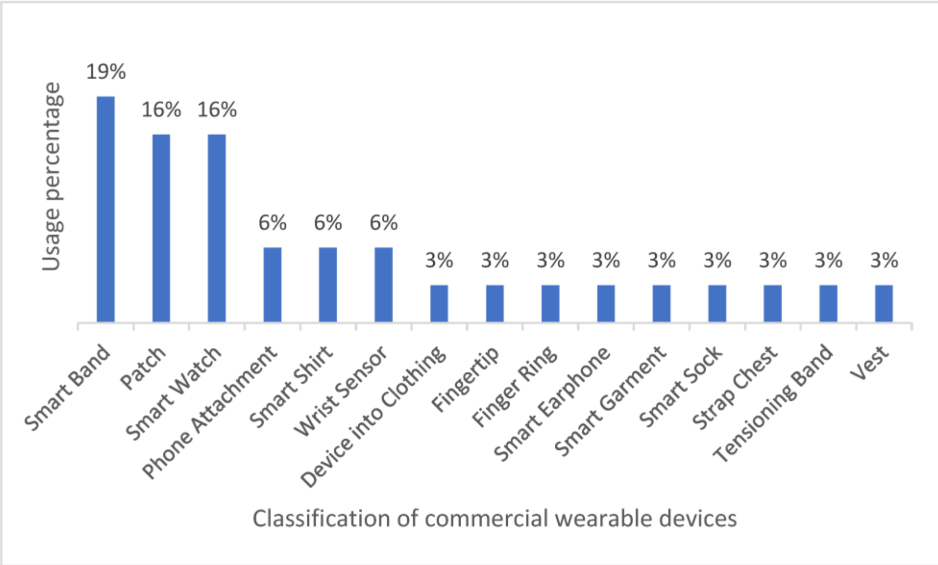


Figure 8: Classification of commercial wearable devices (Prieto-Avalos et al., 2022).

(Paper 4) Huda et al. (2020) proposes a $43 wearable Electrocardiogram system using fabric electrodes that transmits data via Bluetooth to a cloud deep learning model classifying 13 arrhythmia types plus normal sinus rhythm, as shown in Figure 9. Evaluation is quantitative using sensitivity, specificity and F1-score on a benchmark dataset. This is a good example of how end-to-end wearable monitoring can be built affordably by combining low cost hardware with cloud deep learning, which signifies a crucial direction for making these systems accessible in low resource settings.

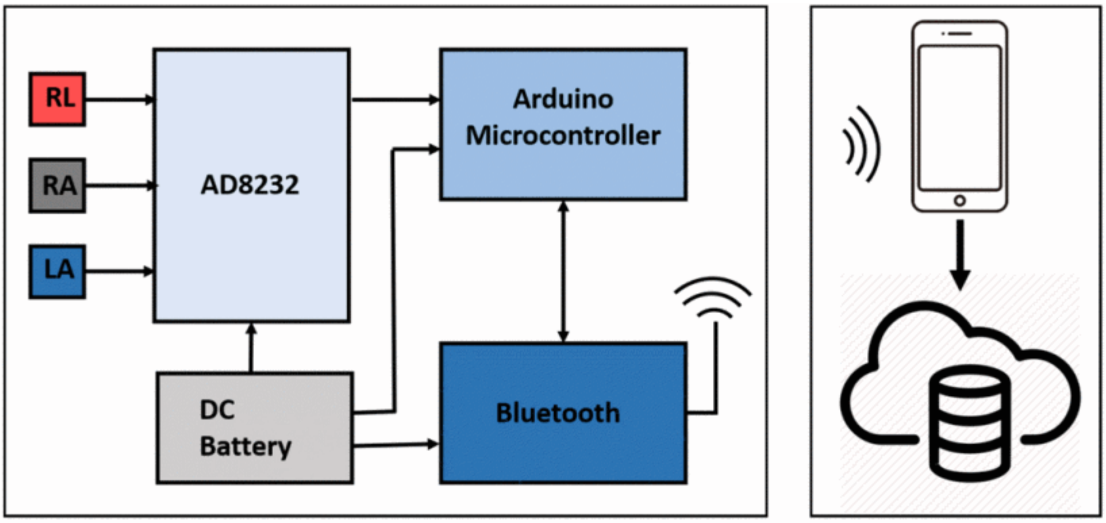


Figure 9: Schematic diagram of the proposed low-cost wearable ECG system (Huda et al., 2020).

2.3 Category C: Mental Health and Sleep Monitoring

(Paper 5) Gedam and Paul (2021) reviews physiological signals from wearables for mental stress detection, comparing Support Vector Machine, random forests, K-Nearest Neighbour and deep neural networks across multiple benchmark datasets. Figure 10 shows common sensor placements on the human body which helps understand how these signals are collected in practice. The systematic review finds that Electrodermal Activity and heart rate are the most discriminative features for stress detection, which signifies how important perticular biosignals are compared to others. On the other hand the paper also proposes a multimodal deep learning stress detection architecture as a contribution.

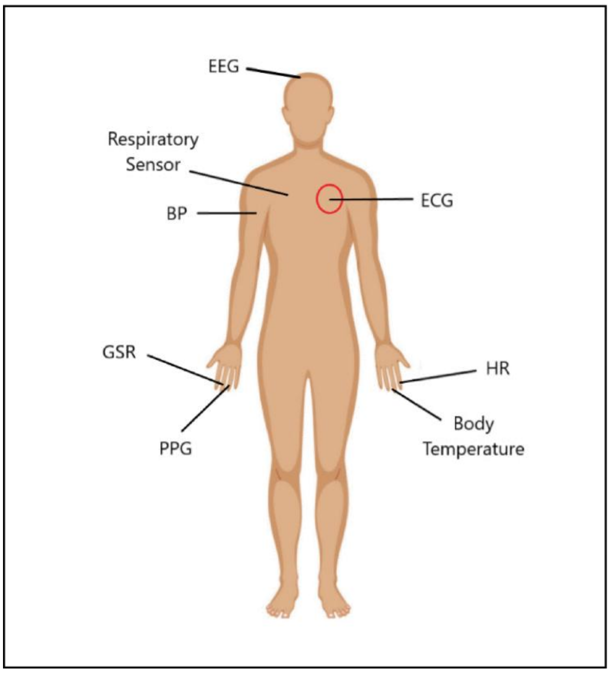


Figure 10: Common wearable sensor placements on the human body (Gedam & Paul, 2021).

(Paper 6) Radha et al. (2021) addresses the scarcity of labelled Photoplethysmography sleep data by applying transfer learning training a deep Recurrent Neural Network on a large Electrocardiogram sleep dataset of 292 participants then transferring weights to a smaller Photoplethysmography dataset of 60 participants for 4 class sleep stage classification, as shown in Figure 11. Quantitative evaluation achieves Cohen's kappa of 0.65 and accuracy of 76.36% which was unprecedented for Photoplethysmography based sleep staging at time of publication. To my knowledge i believe transfer learning is a very useful approach when labelled wearable data is scarce, which is a common situaution across many healthcare applications.

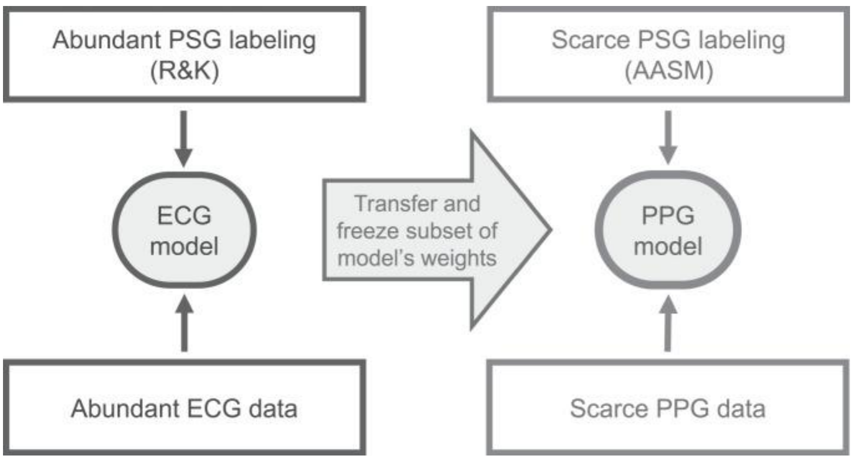


Figure 11: Transfer learning approach (Radha et al., 2021).

2.4 Category D: Activity Recognition and Fall Detection

(Paper 7) Li et al. (2020) proposes a multilayer Bidirectional Long Short-Term Memory framework for continuous activity stream classification and fall detection, combining Frequency Modulated Continuous Wave radar and three wearable Inertial Measurement Unit sensors placed on wrist, waist and ankle as shown in Figure 12. A novel hybrid soft hard fusion scheme is introduced which achieves approximately 96% accuracy on continuous activity streams including falls, outperforming single-modality approaches which signifies the major benefit of combining multiple sensor sources together.

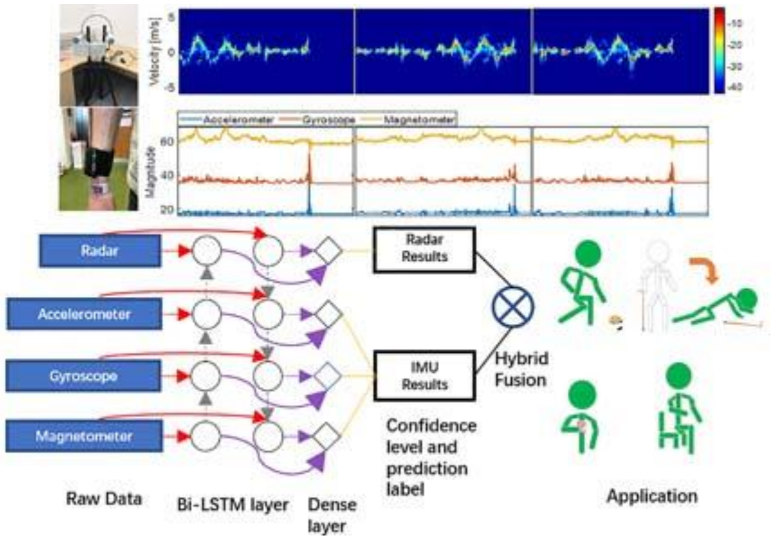


Figure 12: Sketch of the Bi-LSTM based radar and IMU fusion system (Li et al., 2020).

(Paper 8) Mohammad et al., (2023) presents a class based Convolutional Neural Network and Recurrent Neural Network ensemble for elderly fall detection with distinct models per class Non-Fall, Pre-Fall and Fall. Enabling pre-impact warning as shown in Figure 13. Convolutional Neural Network extracts spatial features from accelerometer and gyroscope data while Recurrent Neural Network models the temporal dynamics of the falling process. Quantitative evaluation on the SisFall dataset achieves 95%, 96% and 98% accuracy per class respectively, outperforming prior state-of-the-art methods which is a decent result for this perticular task.

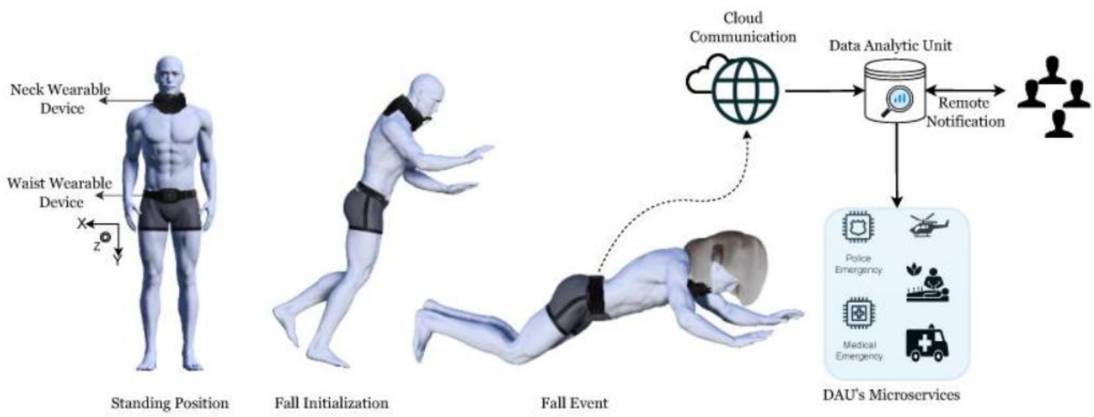


Figure 13: Proposed elderly fall detection framework with CNN-RNN ensemble architecture (Mohammad et al., 2023).

2.5 Category E: Remote Patient Monitoring

(Paper 9) Shaik et al. (2023) reviews Artificial Intelligence enabled remote patient monitoring covering the full architecture from Internet of Things wearables through edge and cloud computing to clinical decisions, as illustrated in Figure 14. Applications include chronic disease management, elderly home care and post hospitalisation monitoring. A structured review addressing four research questions identifies data privacy, device heterogeneity and class imbalance as major unsolved challenges, which signifies how much work is still needed before these systems can be fully trusted in real clinical environments.

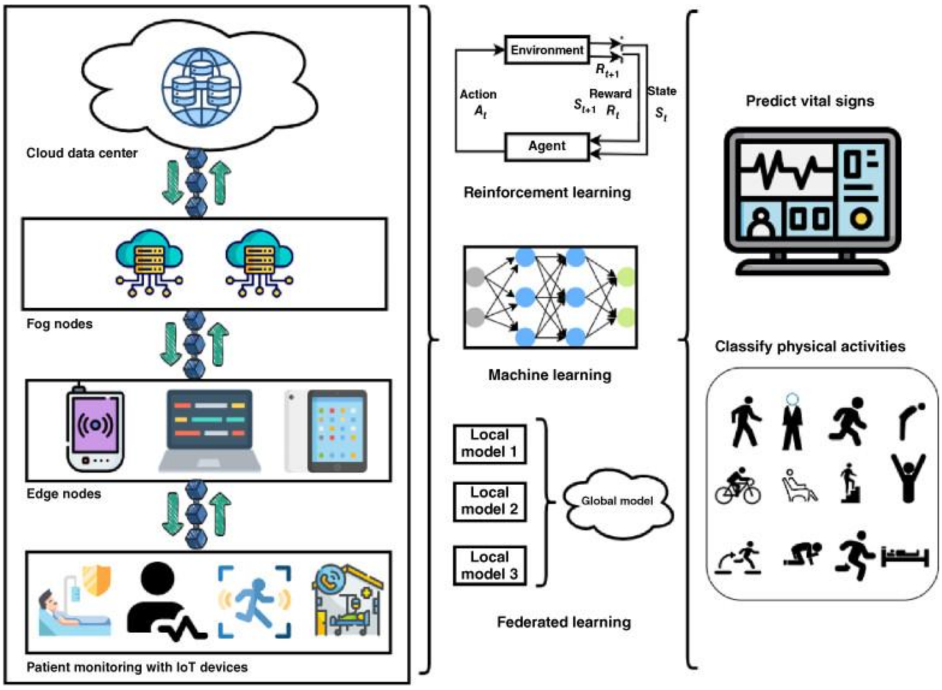


Figure 14: AI-enabled remote patient monitoring architectures (Shaik et al., 2023).

(Paper 10) Islam et al. (2023) builds a complete Internet of Things health monitoring system using three wearable sensors MAX30100 for heart rate and blood oxygen saturation, AD8232 for Electrocardiogram and MLX90614 for body temperature transmitting via Message Queuing Telemetry Transport protocol to a Convolutional Neural Network server for real time abnormality detection and physician alerts, as shown in Figure 15. Quantitative evaluation demonstrates high classification accuracy in a real home monitoring scenario. On the other hand this paper is a good practical example of how machine learning and Internet of Things sensing can combine into a deployed clinical system, which signifies why this research field is so significant and meaningful.

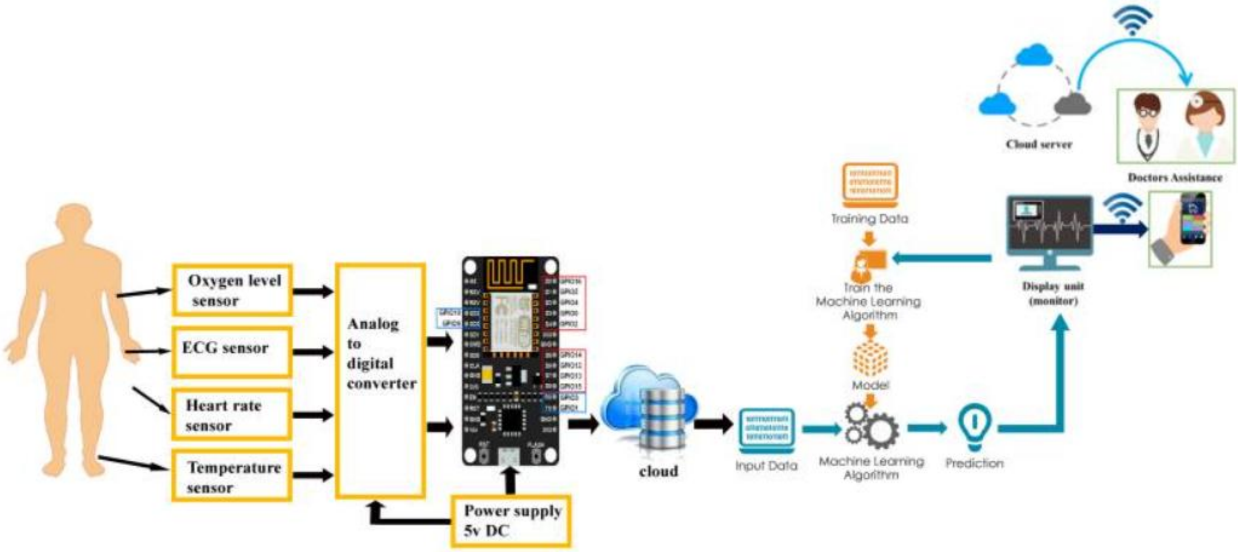


Figure 15: Proposed framework for remote health monitoring using IoT sensors and deep learning ([Islam et al., 2023](#)).

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